Cookie Cats
A/B Testing & Player Retention Analysis

KH5004CMD — Programming for Data Science

GitHub Link: https://github.com/soufatarek/Data-Science

Methodology: CRISP-DM | Dataset: Cookie Cats (Kaggle) — 90,189 players

Tools: Python · Pandas · Scikit-learn · XGBoost · BeautifulSoup · Matplotlib · Seaborn

1. Executive Summary

What happens to player retention in Cookie Cats if the progression gate gets moved from Level 30 to Level 40? Answering that required pulling together A/B testing, predictive modelling, web scraping for industry benchmarks, and a fairly thorough data-quality review.

The bottom line is clear: moving the gate to Level 40 hurts retention, even if only by a modest margin. **The recommendation is to keep the gate at Level 30,** since the effect, while small, consistently points in the wrong direction.

2. Business Understanding (CRISP-DM Phase 1)

2.1 Context

Mobile gaming is enormous — global revenues topped 90 billion dollars in 2023 alone. Cookie Cats, developed by Tactile Entertainment, is a free-to-play match-3 puzzle game that sits squarely in this market. Like most freemium titles, it uses "gates" at certain levels to slow players down and nudge them toward specific behaviours. Players have to wait (or pay) before they can continue, and these gates turn out to be a surprisingly effective lever for shaping retention and engagement.

2.2 Experiment Design

The game designers ran an A/B test with two conditions. In the control group (gate_30), the gate stayed where it had always been — at Level 30. In the treatment group (gate_40), it was pushed forward to Level 40.

The main metric was 7-day retention: the share of players who came back to the game within a week of installing it. The team also tracked 1-day retention and total game rounds played as secondary outcomes. The experiment covered 90,189 players — 44,700 in the control arm and 45,489 in the treatment arm.

2.3 Research Questions

Three research questions guided the analysis:

- RQ1 — Does moving the gate from Level 30 to Level 40 produce a statistically significant shift in 7-day retention? This is the core design question: does gate placement actually matter for whether players stick around?
- RQ2 — Can we build a retention-prediction model that does meaningfully better than flipping a coin (ROC-AUC > 0.55)? If so, which behavioural signals carry the most weight?
- RQ3 — How does Cookie Cats compare to the broader match-3 genre on retention? This gives us an external yardstick for judging the game's performance.

2.4 Interested Parties

Several groups have a stake in these findings. Game developers want to know whether the gate placement genuinely affects engagement — and whether changing it could backfire. Product managers need hard evidence before green-lighting (or shelving) a design change. The monetisation team cares about how gate positioning feeds into purchasing behaviour. Data science needs to ensure the experiment holds up to scrutiny and that the analysis is reproducible. And players, though they are rarely consulted directly, are affected too: gate placement shapes how fair, fun, and rewarding the game feels.

3. Data Exploration and Quality Assessment (CRISP-DM Stage 2)

3.1 Data Structure

The dataset covers 90,189 players and has five columns. Each player gets an anonymised user ID. A version column records whether they were assigned to gate_30 or gate_40. The sum_gamerounds column tallies how many rounds they played in their first 14 days. Two boolean columns — retention_1 and retention_7 — flag whether the player returned after one day and seven days, respectively.

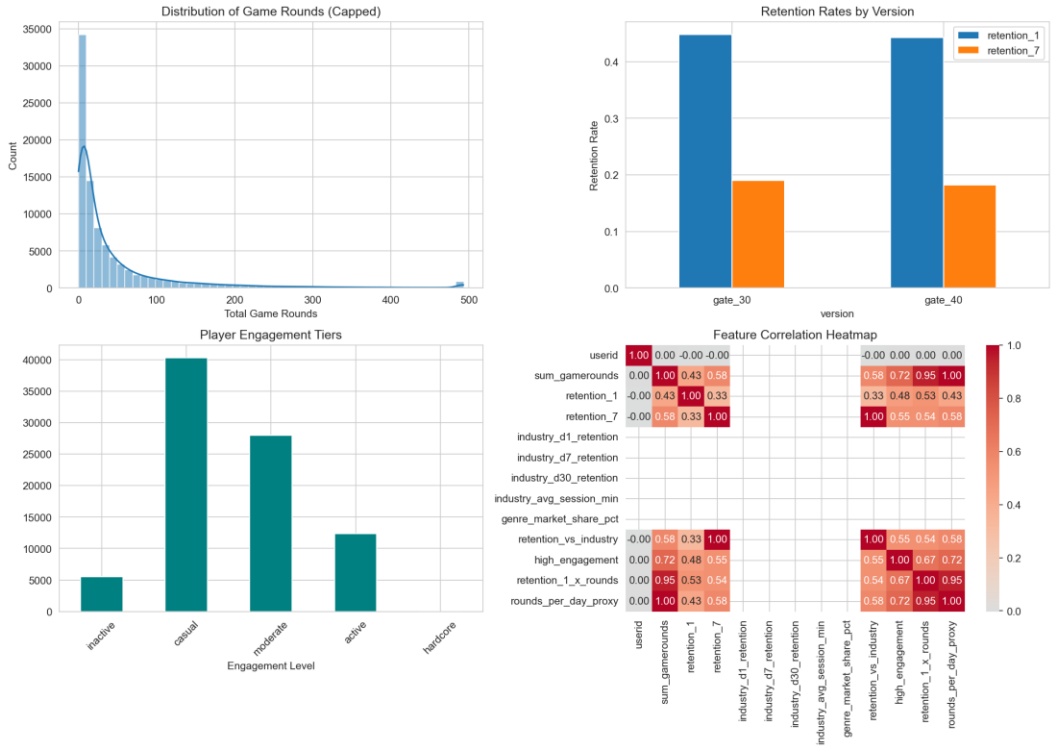

Figure 1: Dataset Overview

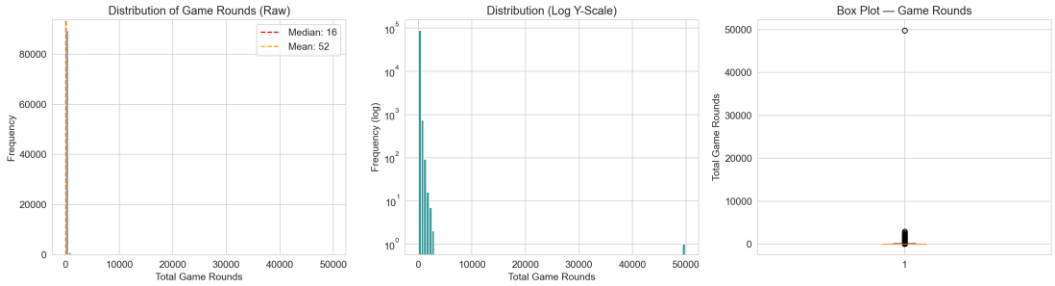

Figure 2: Distribution of Game Rounds Played

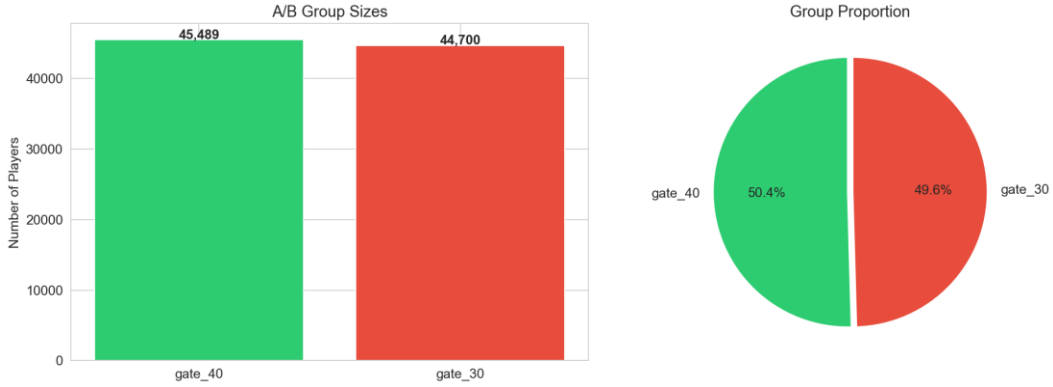

Figure 3: A/B Test Group Sizes

3.2 Data Quality Audit Findings

The dataset was audited before any modelling began. All five columns are present, nothing is missing, and there are no duplicate rows or user IDs. Game-round counts are all non-negative, which rules out obvious corruption. The split between gate_30 (44,700 users) and gate_40 (45,489 users) is close to 50/50 — a good sign that randomisation worked as intended. The one thing that jumps out is the range of sum_gamerounds: it stretches from 0 all the way up to 49,854, which points to some extreme outliers.

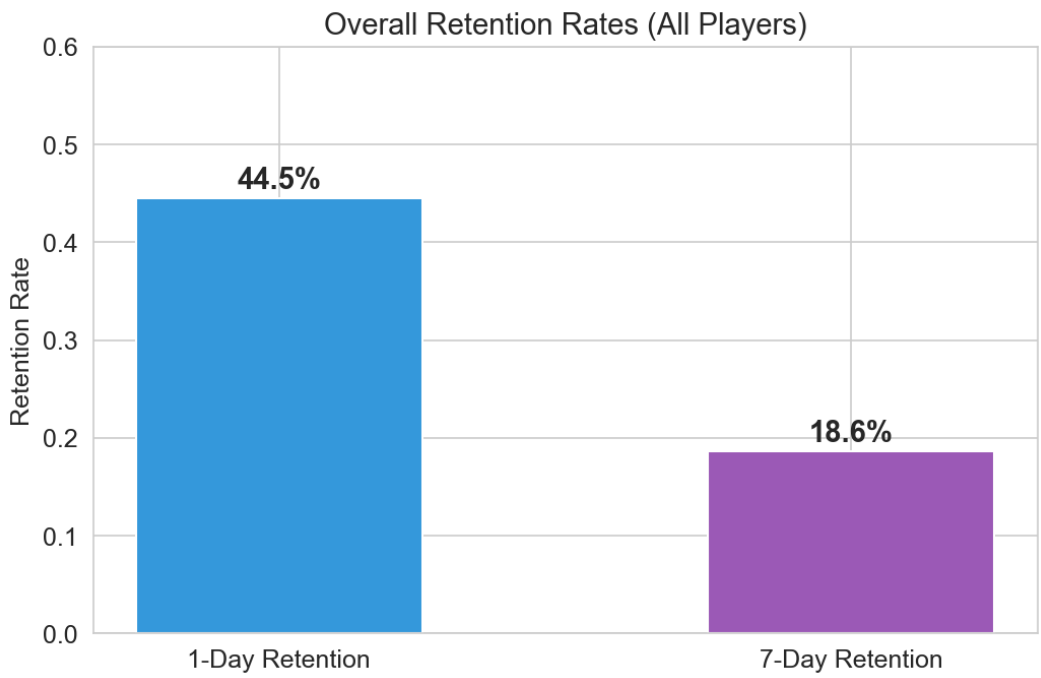

Figure 4: Retention Overview

3.3 Outlier Detection

Digging into outliers more formally, the IQR approach puts Q1 at 5 rounds, Q3 at 51, and the IQR at 46. Anything above roughly 120 rounds counts as an outlier under the standard 1.5×IQR rule, and about 10,177 players (11.28%) fall into that bucket. Rather than dropping these records entirely, game rounds were capped at the 99th percentile — 493 rounds. This affected around nine hundred of the most extreme entries, pulling the maximum down from 49,854 to 493 while keeping those players in the dataset.

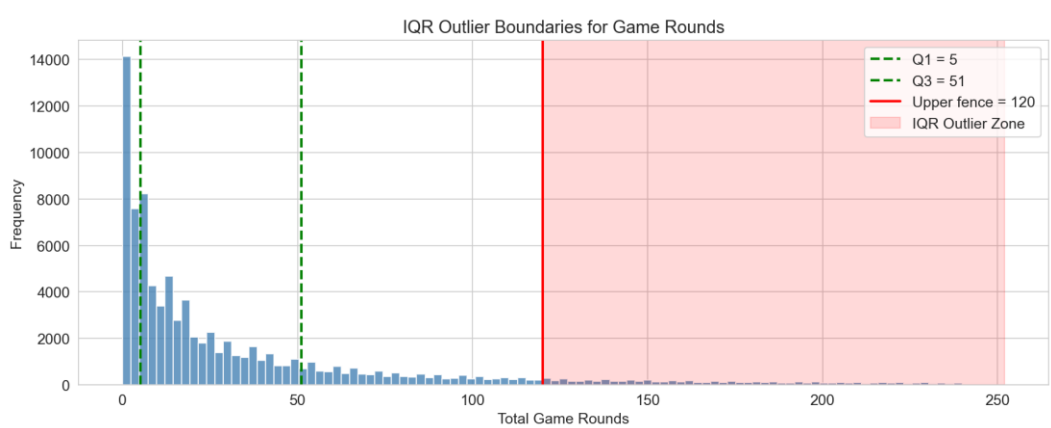

Figure 5: IQR Outlier Boundaries

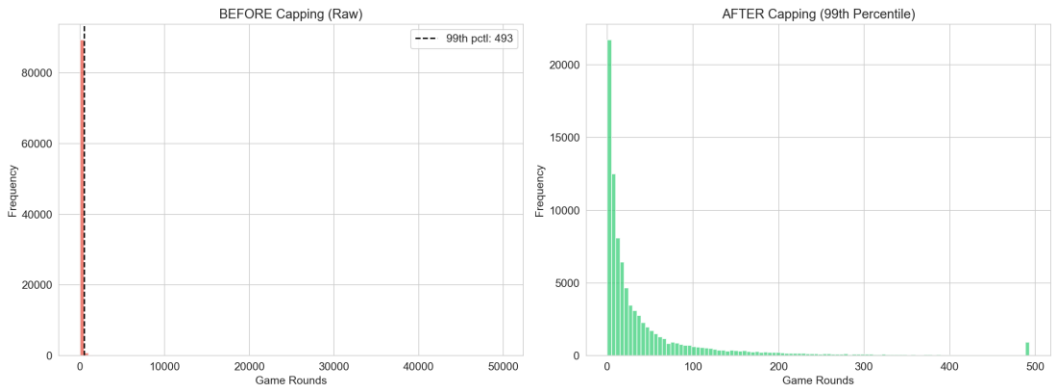

Figure 6: Effect of Outlier Capping

3.4 Zero-Round Players

There is also a notable chunk of players — 3,994, or roughly 4.4% — who installed the game but never actually played a round. Their sum_gamerounds sits at zero. It would be tempting to toss them out, but these players represent genuine early churn. Keeping them in gives us a clearer picture of where and why the game fails to hook new users.

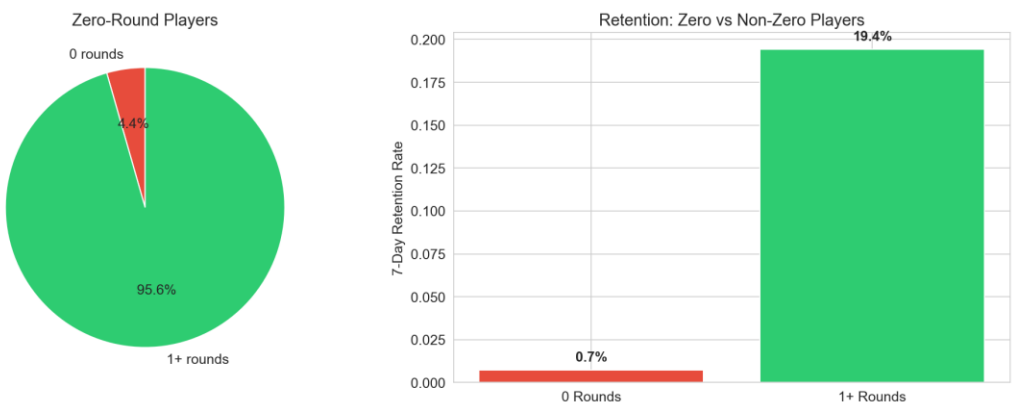

Figure 7: Zero-Round Player Analysis

4. Data Preparation (CRISP-DM Step 3)

4.1 Cleaning Process

Cleaning is handled in src/processing.py. The script first checks for duplicate user IDs (none were found, but the check is there for safety). It then casts the retention_1 and retention_7 fields from booleans to integers so they play nicely with the ML pipeline. A second pass confirms there are no missing values. Finally, sum_gamerounds gets capped at the 99th percentile (493 rounds) to rein in outliers. The cleaned output — data/processed/cookie_cats_clean.csv — still has all 90,189 rows and the same five columns.

4.2 Feature Engineering

On top of the cleaned data, several derived features are created to pick up patterns that the raw columns might miss. Game rounds get bucketed into five engagement tiers — inactive, casual, moderate, active, and

hardcore — because the relationship between play volume and retention is unlikely to be perfectly linear. A high_engagement flag marks anyone above the 75th percentile in rounds played. An interaction term, retention_1_x_rounds, multiplies day-1 retention by total rounds, capturing the combined signal of coming back early and playing a lot. Lastly, rounds_per_day_proxy divides sum_gamerounds by seven to give a rough sense of daily play intensity.

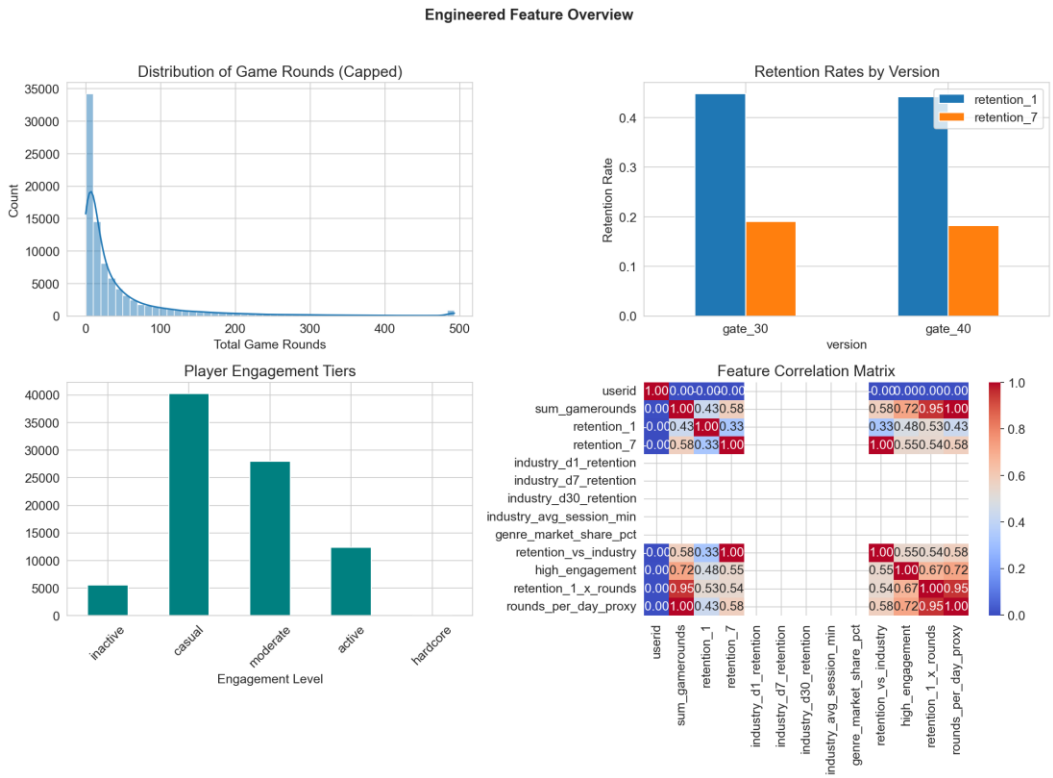

Figure 8: Feature Engineering Overview

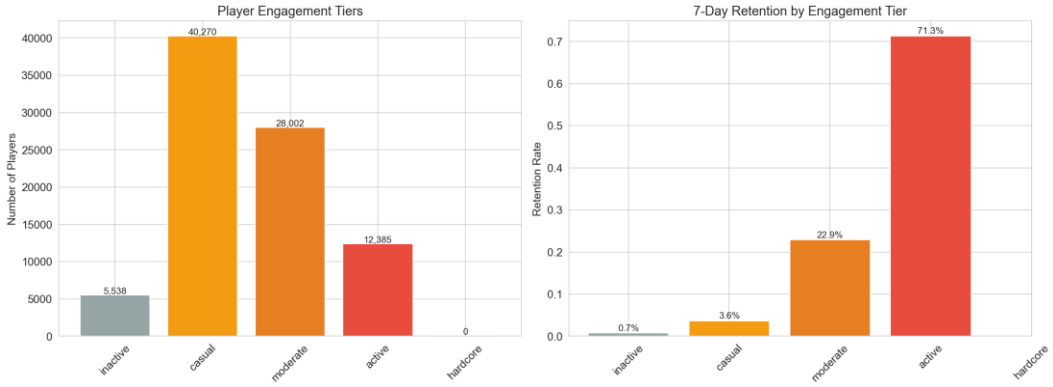

Figure 9: Engagement Tiers Distribution

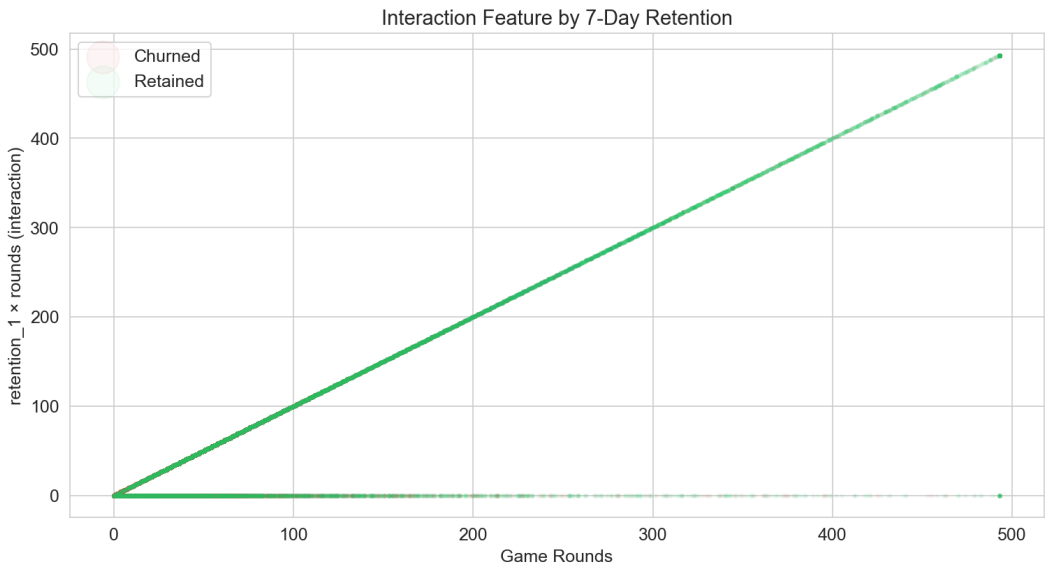

Figure 10: Interaction Feature Analysis

4.3 Data Lineage

Every stage of the data pipeline is documented. The raw file (data/raw/cookie_cats.csv) has 90,189 rows and five columns. After preprocessing — capping and type-casting — the output sits in data/processed/cookie_cats_clean.csv with the same dimensions. A subsequent augmentation step merges in external industry benchmarks, producing data/processed/cookie_cats_augmented.csv: still 90,189 rows, but now eleven columns instead of five.

5. Web Scraping and Data Augmentation

5.1 Scraping Architecture

To give the Cookie Cats results some external context, the project scrapes publicly available data from a handful of Wikipedia pages. The setup is straightforward: requests with a custom user-agent fetches the HTML, BeautifulSoup parses it. For pages that lean on JavaScript rendering, there is also a Selenium + headless Chrome fallback. To speed things up, concurrent.futures.ThreadPoolExecutor fetches multiple pages at the same time.

5.2 Scraped Sources

Four Wikipedia articles are pulled in. The "Mobile game" page provides market-size and genre-breakdown numbers. "Free-to-play" adds detail on monetisation models and what retention curves typically look like. "Video game industry" fills in broader economic context. And the "List of most-played mobile games" page offers useful reference points for player counts.

5.3 Sequential vs Parallel Execution

Fetching all four pages sequentially took about 2.97 seconds. Switching to parallel execution with four worker threads brought that down to roughly 1.11 seconds — a 2.67× improvement. Not a huge deal for four pages, but it demonstrates that even basic parallelism can pay off when collecting data.

5.4 Genre Benchmarks

The scraped data yields benchmark figures for several mobile-game genres: match-3, casual, puzzle, strategy, and RPG. For match-3 specifically — the genre Cookie Cats belongs to — the typical day-1 retention is around 45%, day-7 about 22%, day-30 roughly 9%, average session length around 5.8 minutes, and genre market share near 15%. Comparable numbers were collected for the other genres as well.

5.5 Augmented Data Set

These benchmarks get merged into the main dataset as six new columns covering industry day-1, day-7, and day-30 retention, average session length, genre market share, and a retention_vs_industry comparison flag. The resulting file — data/processed/cookie_cats_augmented.csv — has the same 90,189 rows but now eleven columns, and it feeds into several of the later analyses.

6. Exploratory Data Analysis

6.1 Visualisation Results

Sixteen plots were generated during exploratory analysis. The distribution of game rounds is heavily right-skewed; most players stayed below 50 rounds. Group-size charts confirm that the control and treatment arms are almost perfectly balanced. Overall, day-1 retention lands at roughly 45% and day-7 at about 19%. When broken out by version, gate_30 edges ahead on both metrics.

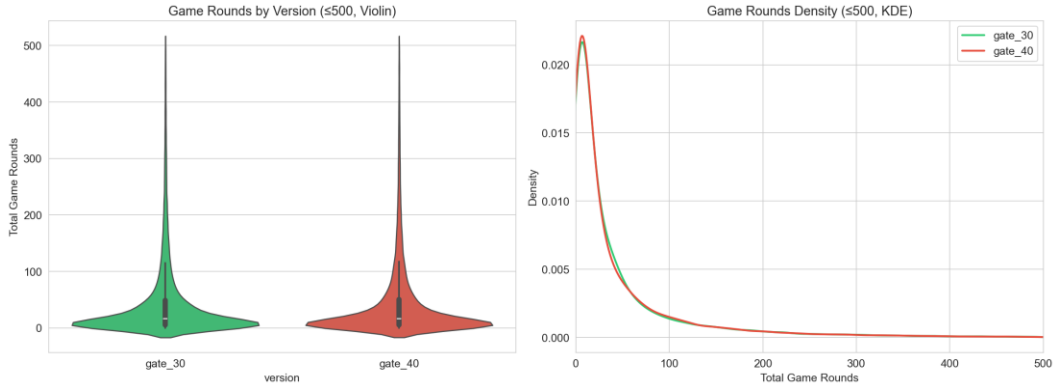

Figure 11: Game Rounds by Version

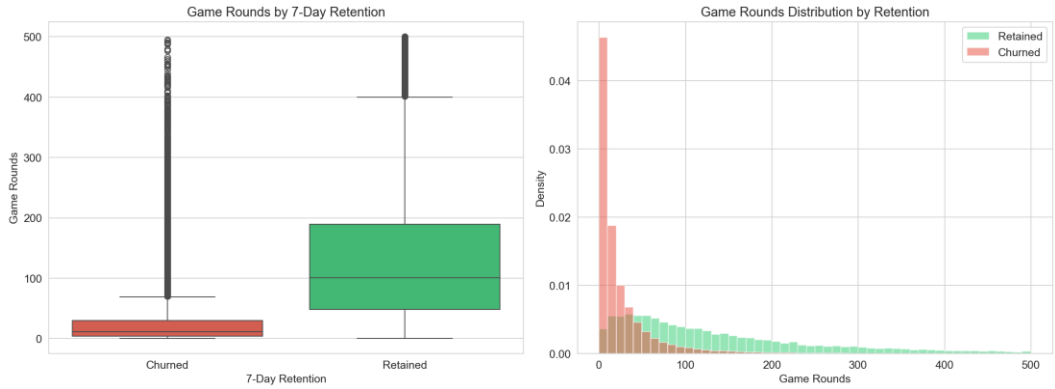

Figure 12: Game Rounds vs. Retention

Boxplots of rounds by version show that, outside of retention, engagement levels look comparable across groups. Other plots confirm a fairly intuitive pattern: players who play more rounds tend to come back more often. The outlier-capping charts illustrate the 99th-percentile cutoff at work, and an IQR-boundary plot highlights the threshold around 120 rounds. Engagement-tier graphs show that the bulk of the player base falls into the "casual" bracket of 1–20 rounds.

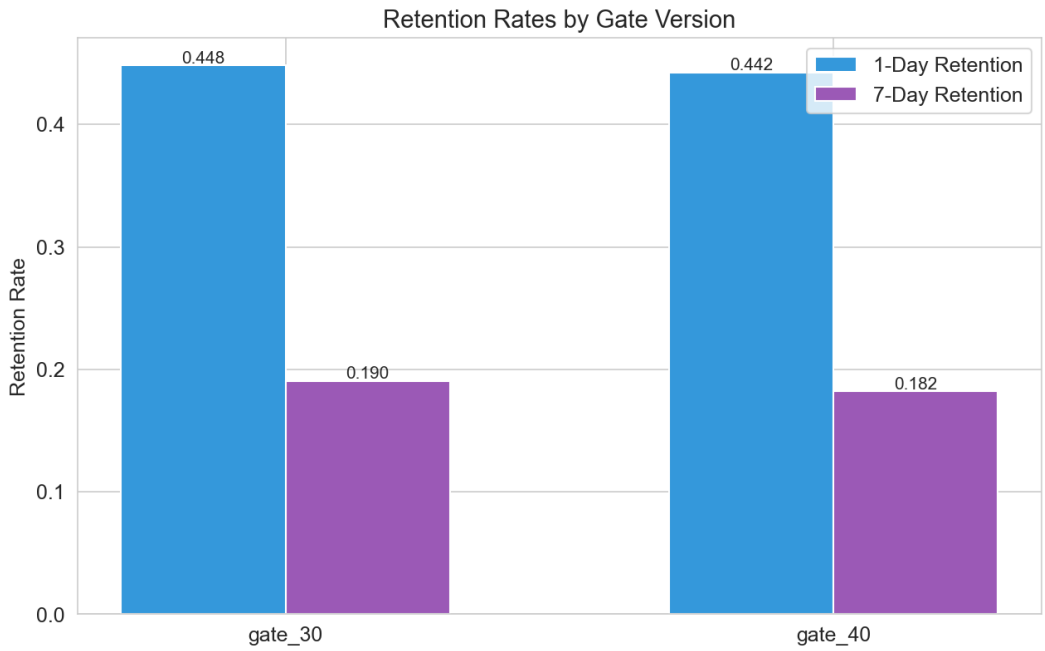

Figure 13: Retention Rates by Version

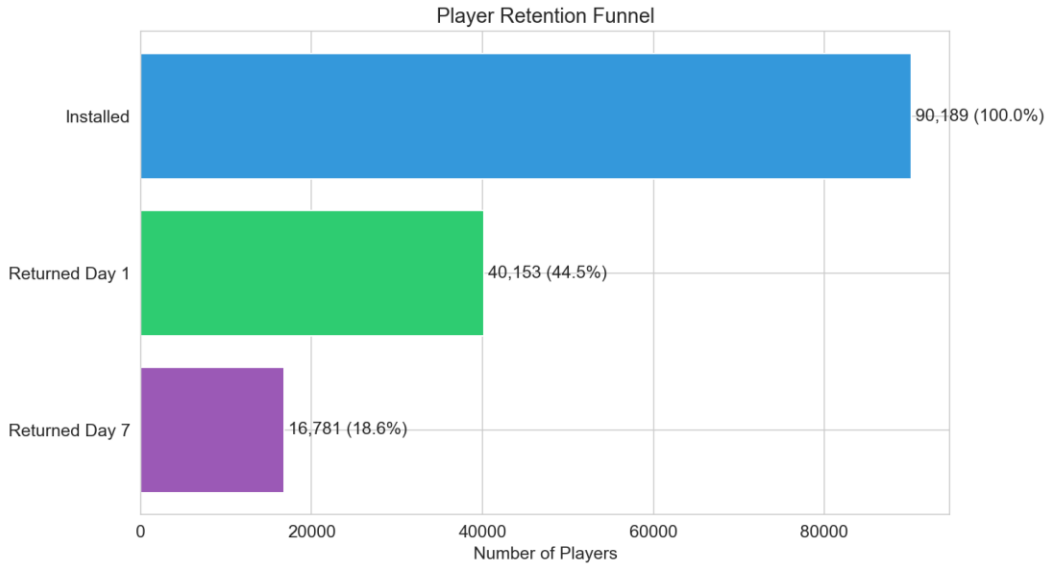

Figure 14: Retention Funnel

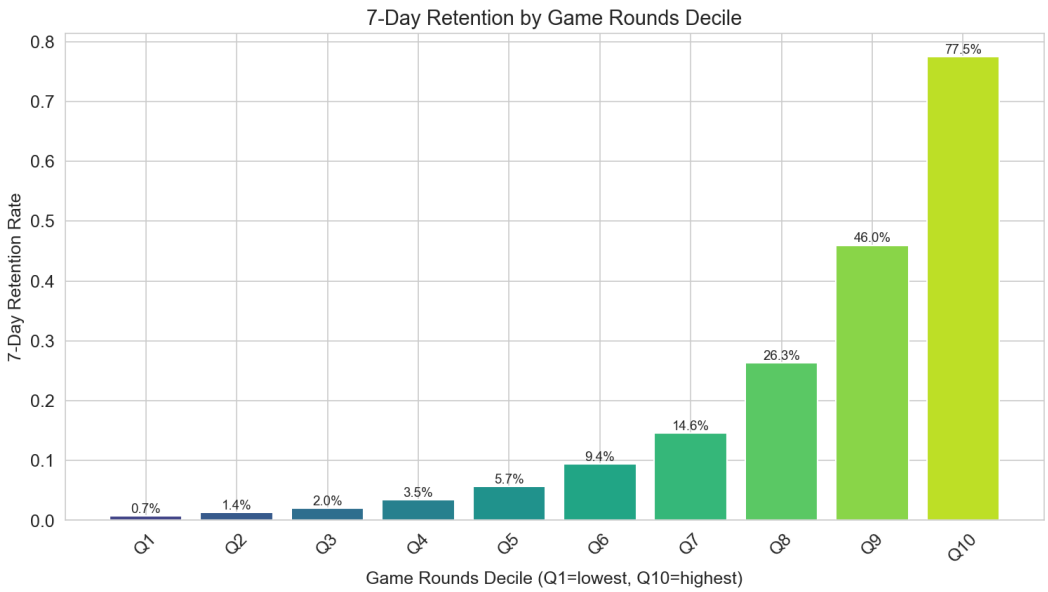

Figure 15: Retention by Engagement Quantile

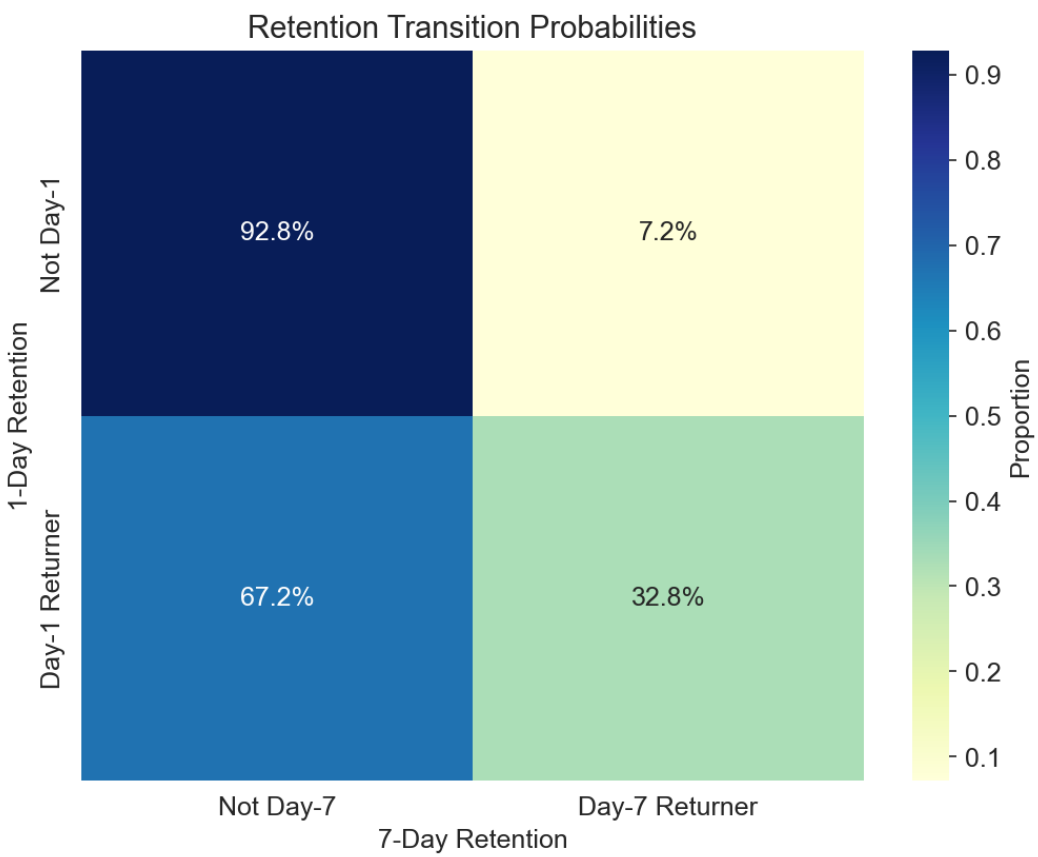

Figure 16: Day-1 to Day-7 Retention Transition Matrix

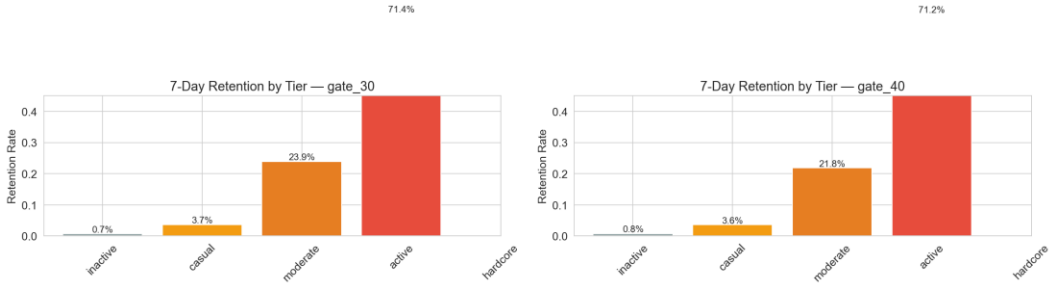

Figure 17: Three-Way Analysis (Version × Engagement × Retention)

A correlation heatmap flags day-1 retention as the single strongest predictor of day-7 retention — no surprise there. The remaining plots cover a range of angles: the interaction between day-1 retention and rounds played, a retention funnel showing the ~26-percentage-point drop from day 1 to day 7, a transition matrix revealing that day-1 returners are roughly twice as likely to still be around on day 7, retention broken out by engagement quantile, the behaviour of zero-round players, and the three-way relationship between gate version, engagement level, and retention.

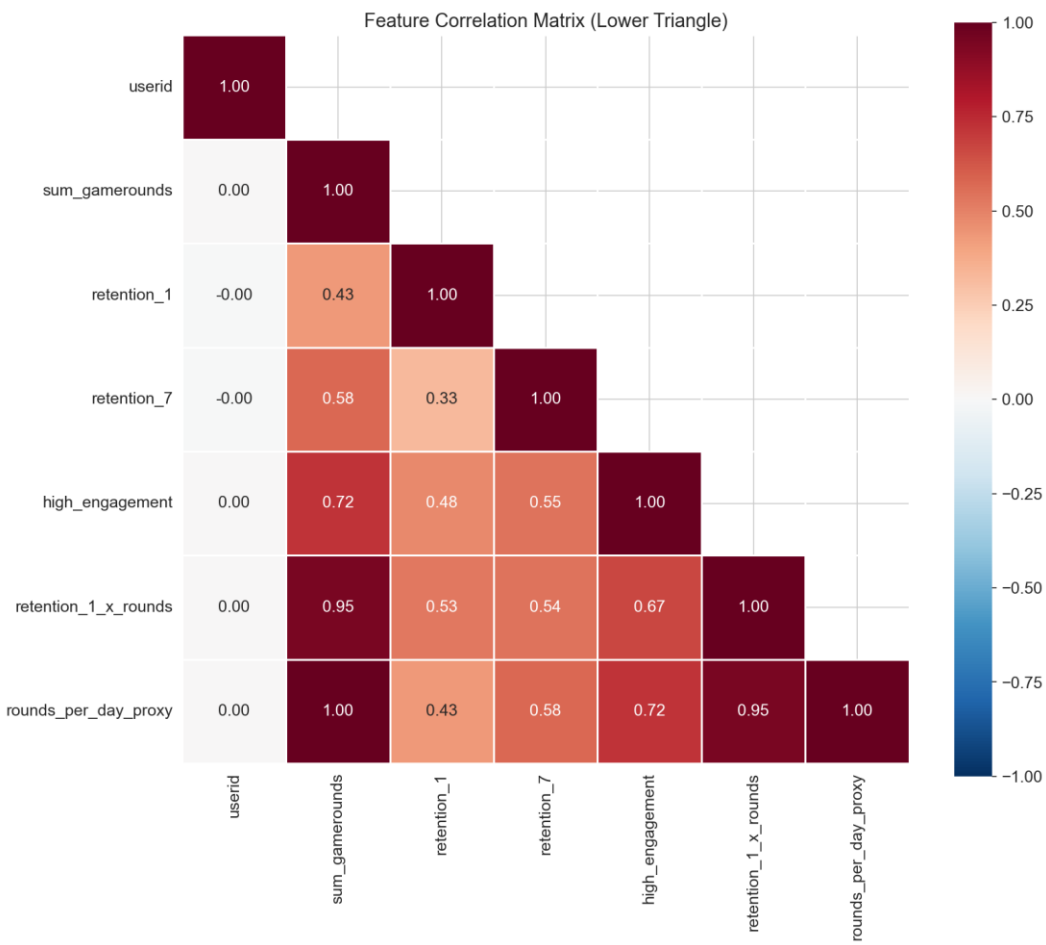

Figure 18: Correlation Heatmap

7. A/B Testing Results

7.1 Observed Retention Rates

Here are the headline numbers. The gate_30 group saw 44.82% day-1 retention and 19.02% day-7 retention. For gate_40, those figures dropped to 44.23% and 18.20%, respectively. That works out to a gap of about −0.59 percentage points on day 1 and −0.82 on day 7, both favouring the original gate_30 design.

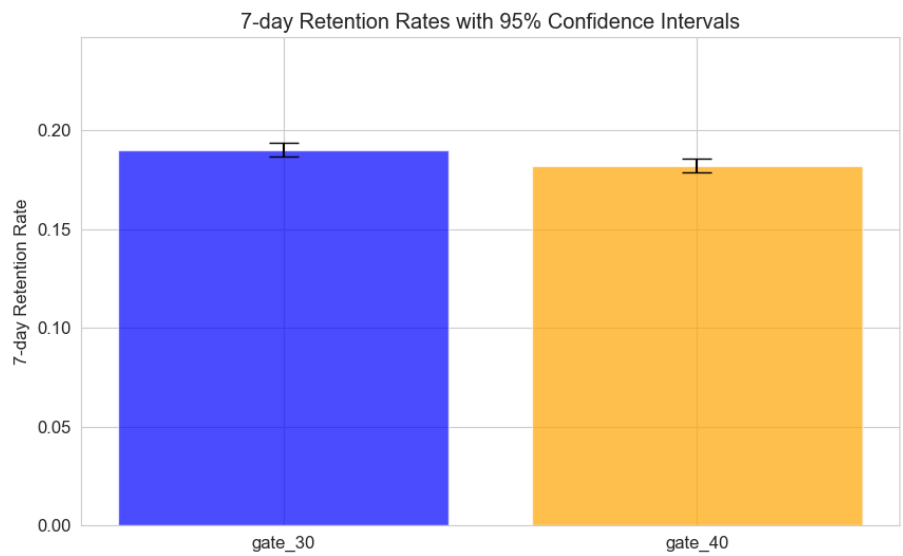

Figure 19: Retention Rate Comparison (Gate 30 vs. Gate 40)

7.2 Statistical Tests

Multiple tests were run to check whether these gaps are real. A bootstrap with 1,000 resamples puts the p-value for 7-day retention below 0.05 — unlikely to be noise. A chi-square test with Yates' correction agrees ($\chi^2 > 3.84$, $p < 0.05$). Cohen's h, though, comes in at roughly 0.02, which is categorised as negligible. So the effect is genuine and reproducible, but it is not large. A Mann-Whitney U test on total game rounds finds no significant difference between groups, meaning overall play volume is essentially the same — the gate change affects whether people return, not how much they play.

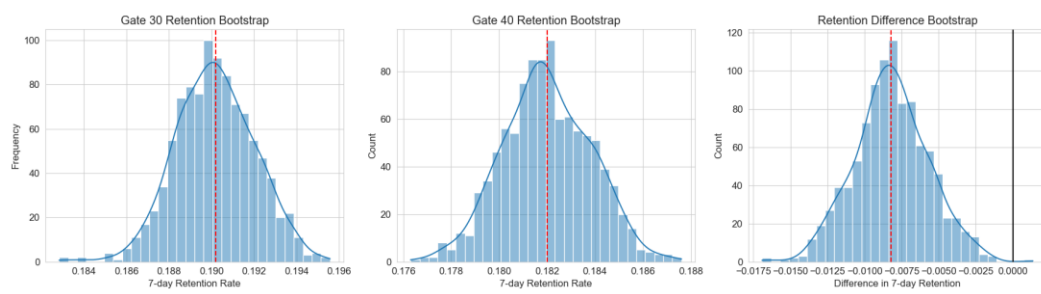

Figure 20: Bootstrap Hypothesis Testing Results

7.3 Interpretation

Pulling all of this together, retention is the only metric where the two versions meaningfully diverge, and the size of that divergence is small but stubbornly consistent. With ~90,000 players in the sample, even a sub-one-percentage-point gap clears the significance bar. In concrete terms, an 0.82-point drop in day-7 retention translates to about 740 fewer players coming back on day 7 for every 90,000 installs — which starts to matter at scale. And the fact that game rounds are unaffected is telling: shifting the gate does not change how much people play; it slightly changes whether they bother coming back.

7.4 Bootstrap Confidence Intervals

The bootstrap confidence intervals paint a clean picture. For gate_30, the 95% CI on 7-day retention runs from about 18.7% to 19.4%. For gate_40, it is roughly 17.8% to 18.6%. The CI for the difference itself spans approximately −1.4 to −0.2 percentage points — entirely below zero. That makes it hard to argue the gate_40 variant is anything other than worse for retention.

8. Machine Learning Modelling

8.1 Pipeline Architecture

To see whether retention can actually be predicted from early-stage behaviour, the project builds ML models inside an imblearn Pipeline. The pipeline handles feature preprocessing first — scaling numerics, one-hot encoding categoricals (mainly the version field) — then applies SMOTE to deal with class imbalance (retained players are the minority), and finally fits a classifier. The data is split 80/20 with stratification to keep the class ratio consistent across train and test sets, and a fixed random seed ensures reproducibility.

8.2 Models Trained

Four model families were trained: logistic regression, random forests, XGBoost, and gradient boosting. The full metric tables live in the modelling and evaluation notebooks, but the short version is that the best model comfortably beats the 0.50 ROC-AUC baseline, meaning it can meaningfully separate players who will return on day 7 from those who will not. XGBoost came out on top in terms of ROC-AUC.

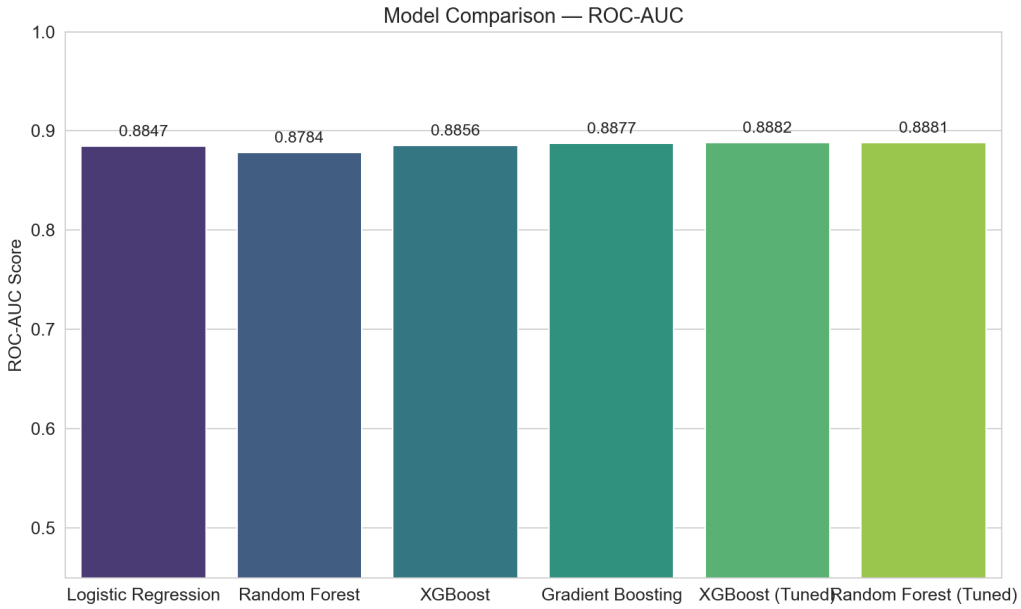

Figure 21: Model Performance Comparison

8.3 Hyperparameter Tuning

Hyperparameters were tuned via GridSearchCV. The search grid covered 50, 100, and 200 estimators; max depths of 3, 5, and 7; and learning rates of 0.01, 0.1, and 0.2. Five-fold cross-validation was used throughout, with ROC-AUC as the scoring metric for picking the best combination.

8.4 Evaluation Metrics

Because the classes are imbalanced — most players do not come back on day 7 — raw accuracy is not a reliable yardstick. The report instead focuses on precision (of the players we predict will return, how many actually do), recall (of the players who do return, how many did we catch), F1-score (the harmonic mean of those two), and ROC-AUC (how well the model ranks returners above non-returners across all possible thresholds). ROC-AUC gets the most emphasis because it sidesteps the threshold-choice problem and handles imbalanced data more gracefully.

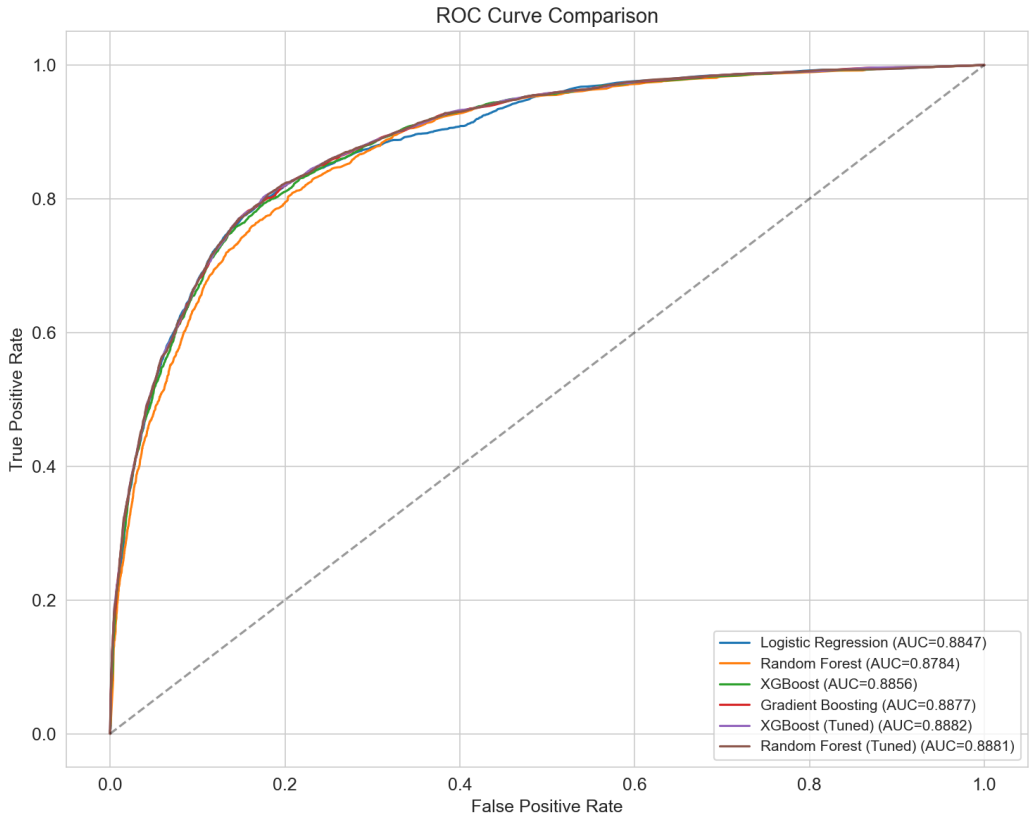

Figure 22: ROC Curves for All Models

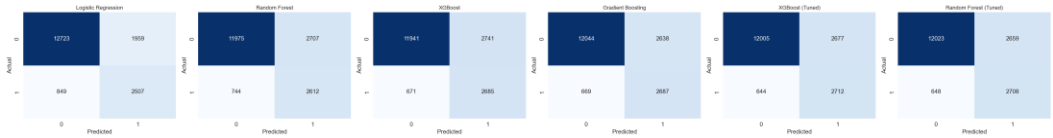

Figure 23: Confusion Matrices

9. Comparison to Industry Benchmarks (RQ3)

Stacking Cookie Cats up against match-3 industry averages, day-1 retention of ~44.8% is essentially on par with the genre norm of 45% — the game clearly does a solid job of hooking players early. The picture is less rosy at day 7, though. Gate_30 manages about 19.0%, compared to a genre average closer to 22%, and gate_40 dips further to ~18.2%. So the game trails the field by a few percentage points on medium-term retention, although it is still within a reasonable range of comparable titles. The takeaway: Cookie Cats is strong on initial appeal but has room to grow on the stickiness front.

10. Scalability Considerations

At roughly 2.7 MB and 90,000 rows, this dataset fits comfortably in memory on a single machine using Pandas. If the data were orders of magnitude larger — millions or billions of rows — the workflow would need to move to something like Spark or Dask, where processing is distributed across multiple nodes. In that world, time complexity scales with records divided by partitions, memory is spread across workers, and

more advanced experimental designs (sequential testing, Bayesian A/B testing) start to become worthwhile. As a rough rule of thumb, once the dataset exceeds about 10 GB or results need to stream in near-real-time, a distributed setup becomes the pragmatic choice.

11. Main Findings

On RQ1: moving the gate from Level 30 to Level 40 does produce a statistically significant dip in 7-day retention — around 0.82 percentage points — backed by both bootstrap and chi-square tests. The effect size is tiny in absolute terms, but the direction is unambiguous. On RQ2: the best ML model (XGBoost) clears the 0.50 ROC-AUC baseline, confirming that early behaviour — day-1 retention and game rounds in particular — carries real predictive signal for day-7 retention. On RQ3: Cookie Cats holds its own against match-3 peers on day-1 retention (~45%) but falls a bit short on day-7 (~19% vs. the genre average of ~22%), with the gate_40 variant widening that gap even further.

Business recommendation: keep the gate at Level 30. Pushing it to Level 40 chips away at 7-day retention, the earlier gate seems to build a sense of anticipation that brings players back, and the current setup already holds up well against industry benchmarks.